

Copy of Project Overview: The ... internship level 1.ipynb - Colab

Ask Gemini

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Colab logo

internship level 1.ipynb

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Commands + Code + Text Run all

RAM Disk

LEVEL 1 - TASK 1.1

import pandas as pd

df = pd.read_csv("Dataset1.csv", engine='python')

print("\n===== FIRST 5 ROWS =====\n")

print(df.head())

print("\n===== LAST 5 ROWS =====\n")

print(df.tail())

print("\n===== DATASET SHAPE =====\n")

print("Total Records :", df.shape[0])

print("Total Columns :", df.shape[1])

print("\n===== COLUMN NAMES =====\n")

print(df.columns)

print("\n===== DATA TYPES =====\n")

print(df.dtypes)

print("\n===== DATASET INFORMATION =====\n")

df.info()

print("\n===== DESCRIPTIVE STATISTICS =====\n")

print(df.describe())

print("\nTask 1.1 Completed Successfully.")

Variables

Terminal

3:25 PM

Python 3

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First 5 rows

SN	Train No	Station Code	1A	2A	3A	SL	Station Name	Route Number
0	1	107	SWV	100	100	100	SAWANTWADI R	1
1	2	107	THVM	260	228	196	THIVIM	1
2	3	107	KRMT	345	296	247	KARMALI	1
3	4	107	MAO	490	412	334	MADGOAN JN.	1
4	1	108	MAO	100	100	100	MADGOAN JN.	1

Arrival time Departure Time Distance

0	00:00:00	10:25:00	0
1	11:06:00	11:08:00	32
2	11:28:00	11:30:00	49
3	12:10:00	00:00:00	78
4	00:00:00	20:30:00	0

Last 5 rows

SN	Train No	Station Code	1A	2A	3A	SL	Station Name
186069	1	22439	NDLS	100	100	100	NEW DELHI
186070	2	22439	UMB	1095	896	697	AMBALA CANT JN
186071	3	22439	LDH	1660	1348	1036	LUDHIANA JN
186072	4	22439	JAT	2985	2408	1831	JAMMU TAWI
186073	5	22439	SVDK	3375	2720	2065	SHIMLA VD KATRA

Route Number Arrival time Departure Time Distance

186069	1	00:00:00	06:00:00	0
186070	1	08:08:00	08:10:00	199
186071	1	09:19:00	09:21:00	312
186072	1	12:38:00	12:40:00	577
186073	1	14:00:00	00:00:00	655

Variables Terminal

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```
===== DATASET SHAPE =====
'''
Total Records : 186074
Total Columns : 12

===== COLUMN NAMES =====

Index(['SN', 'Train_No', 'Station_Code', '1A', '2A', '3A', 'SL',
       'Station_Name', 'Route_Number', 'Arrival_time', 'Departure_Time',
       'Distance'],
      dtype='object')

===== DATA TYPES =====

SN                int64
Train_No          int64
Station_Code      object
1A                int64
2A                int64
3A                int64
SL                int64
Station_Name      object
Route_Number      int64
Arrival_time      object
Departure_Time    object
Distance          int64
dtype: object

===== DATASET INFORMATION =====

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 186074 entries, 0 to 186073
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -

```

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===== DATASET INFORMATION =====
'''
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 186074 entries, 0 to 186073
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0    SN              186074 non-null  int64
1    Train_No        186074 non-null  int64
2    Station_Code    186074 non-null  object
3    1A              186074 non-null  int64
4    2A              186074 non-null  int64
5    3A              186074 non-null  int64
6    SL              186074 non-null  int64
7    Station_Name    186074 non-null  object
8    Route_Number    186074 non-null  int64
9    Arrival_time    186074 non-null  object
10   Departure_Time  186074 non-null  object
11   Distance        186074 non-null  int64
dtypes: int64(8), object(4)
memory usage: 17.0+ MB

===== DESCRIPTIVE STATISTICS =====

count    SN      Train No      1A      2A \
mean    13.914695  42157.747455  1506.769189  1225.415351
std     12.779368  25090.080221  2418.719821  1934.975856
min      1.000000    107.000000    100.000000    100.000000
25%      5.000000   17225.000000    215.000000    192.000000
50%     11.000000   40050.000000    465.000000    392.000000
75%     18.000000   57550.000000   1555.000000   1264.000000
max     118.000000  99908.000000  21400.000000  17140.000000

3A      SL      Route_Number      Distance
```

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Task 1.1 Completed Successfully.

DESCRIPTIVE STATISTICS

	SN	Train No	1A	2A
count	186074.000000	186074.000000	186074.000000	186074.000000
mean	13.914695	42157.747455	1506.769189	1225.415351
std	12.779368	25090.080221	2418.719821	1934.975856
min	1.000000	107.000000	100.000000	100.000000
25%	5.000000	17225.000000	215.000000	192.000000
50%	11.000000	40050.000000	465.000000	392.000000
75%	18.000000	57550.000000	1555.000000	1264.000000
max	118.000000	99908.000000	21400.000000	17140.000000

	3A	SL	Route Number	Distance
count	186074.000000	186074.000000	186074.0	186074.000000
mean	944.061513	662.735777	1.0	281.353838
std	1451.231892	967.524012	0.0	483.743964
min	100.000000	100.000000	1.0	0.000000
25%	160.000000	146.000000	1.0	23.000000
50%	310.000000	246.000000	1.0	73.000000
75%	973.000000	682.000000	1.0	291.000000
max	12880.000000	8620.000000	1.0	4260.000000

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CommandsCodeTextRun all

LEVEL 1 - TASK 1.2
import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python')

df = df.sort_values(by=["Train_No", "SN"])

train_route = df.groupby("Train_No").agg(
 Start_Station=("Station_Name", "first"),
 End_Station=("Station_Name", "last"),
 Number_of_Stops=("Station_Name", "count")
) .reset_index()

print("\n===== TRAIN-WISE ROUTE TABLE =====\n")
print(train_route)

train_route.to_csv("Train_Wise_Route_Table.csv", index=False)

print("\nTrain-wise route table saved successfully.")

print("\nTask 1.2 Completed Successfully.")

...
===== TRAIN-WISE ROUTE TABLE =====

	Train_No	Start_Station	End_Station	Number_of_Stops
0	107	SAWANTWADI R	MADGOAN JN.	4
1	108	MADGOAN JN.	SAWANTWADI R	4
2	128	MADGOAN JN.	CHHATRAPATI	22
3	290	DELHI-SAFDAR	DELHI-SAFDAR	14
4	401	AURANGABAD	VARANASI JN.	12
...	...	...	...	...

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CommandsCodeTextRun all

===== TRAIN-WISE ROUTE TABLE =====

	Train_No	Start_Station	End_Station	Number_of_Stops
0	107	SAWANTWADI R	MADGOAN JN.	4
1	108	MADGOAN JN.	SAWANTWADI R	4
2	128	MADGOAN JN.	CHHATRAPATI	22
3	290	DELHI - SAFDAR	DELHI - SAFDAR	14
4	401	AURANGABAD	VARANASI JN.	12
...	...	...	...	...
11108	99904	PUNE JN.	TALEGAON	12
11109	99905	TALEGAON	SHIVAJINAGAR	11
11110	99906	PUNE JN.	TALEGAON	12
11111	99907	TALEGAON	PUNE JN.	12
11112	99908	PUNE JN.	TALEGAON	12

[11113 rows x 4 columns]

Train-wise route table saved successfully.

Task 1.2 Completed Successfully.

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Snipping Tool

Screenshot copied to clipboard
Automatically saved to screenshots folder.

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```
# LEVEL 1 - TASK 1.3
import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python', on_bad_lines='skip')

print("\n===== DISTANCE STATISTICS =====\n")
print(df[["Distance"]].describe())

stops = df.groupby("Train_No").size()

print("\n===== NUMBER OF STOPS STATISTICS =====\n")
print(stops.describe())

distance_stats = df[["Distance"]].describe()
stop_stats = stops.describe()

statistics = pd.DataFrame({
    "Distance": distance_stats,
    "Number_of_Stops": stop_stats
})

statistics.to_csv("Basic_Statistics.csv")

print("\nBasic statistics saved successfully.")

print("\nTask 1.3 Completed Successfully.")

...
===== DISTANCE STATISTICS =====
```

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9:29 PM Python 3

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```
===== DISTANCE STATISTICS =====
...
count    186074.000000
mean      281.353838
std       483.743964
min        0.000000
25%       23.000000
50%       73.000000
75%      291.000000
max     4260.000000
Name: Distance, dtype: float64

===== NUMBER OF STOPS STATISTICS =====

count    11113.000000
mean      16.743814
std       12.993123
min        2.000000
25%        8.000000
50%       15.000000
75%       22.000000
max       118.000000
dtype: float64

Basic statistics saved successfully.

Task 1.3 Completed Successfully.
```

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Colab interface showing a Jupyter Notebook with Python code for data quality checks.

Browser tabs: internship level 1(1,2).ipynb - Colab, Ask Gemini

URL: colab.research.google.com/drive/1hW8UZh9mGZeKsu4nUWAv3bPat21tArK#scrollTo=W7lsA3UKj8DK

Code Editor:

```
# LEVEL 1 - TASK 1.4
import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python')

missing_values = df.isnull().sum()

duplicate_records = df.duplicated().sum()

empty_station_names = (df["Station_Name"].astype(str).str.strip() == "").sum()

negative_distance = (df["Distance"] < 0).sum()

summary = pd.DataFrame({
    "Check": [
        "Missing Values",
        "Duplicate Records",
        "Empty Station Names",
        "Negative Distance Values"
    ],
    "Count": [
        missing_values.sum(),
        duplicate_records,
        empty_station_names,
        negative_distance
    ]
})

print("\n===== MISSING VALUES =====\n")
print(missing_values)

print("\n===== DATA QUALITY SUMMARY =====\n")
print(summary)
```

Bottom bar: Variables, Terminal, 9:33 PM, Python 3

internship level 1(1.2).ipynb - Co

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```
summary.to_csv("Data_Quality_Summary.csv", index=False)

print("\nData quality summary saved successfully.")

print("\nTask 1.4 Completed Successfully.")

...

===== MISSING VALUES =====

SN          0
Train_No    0
Station_Code 0
1A          0
2A          0
3A          0
SL          0
Station_Name 0
Route_Number 0
Arrival_time 0
Departure_Time 0
Distance    0
dtype: int64

===== DATA QUALITY SUMMARY =====

      Check  Count
0  Missing Values    0
1  Duplicate Records    0
2  Empty Station Names    0
3  Negative Distance Values    0

Data quality summary saved successfully.

Task 1.4 Completed Successfully.
```

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